

Advanced Features & Power BI Service

From Desktop to Cloud — Publishing, Sharing & Securing Your Reports

Topics Covered in This Chapter

- ▶ Publishing Reports to Power BI Service
 - ▶ Workspaces and Sharing with Your Team
 - ▶ Row-Level Security (RLS) — Controlling Who Sees What
 - ▶ Gateways — Connecting On-Premises Data to the Cloud
 - ▶ Scheduled Refresh — Keeping Your Data Always Up to Date
 - ▶ Bookmarks and Buttons — Interactive Navigation
 - ▶ App Creation — Packaging Reports for End Users
 - ▶ Practice Exercise — End-to-End Deployment Scenario
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Introduction to Power BI Service

Until now, you have been building reports in Power BI Desktop — the free software application installed on your computer. Power BI Desktop is a powerful tool for designing, building, and testing reports. However, it has one significant limitation: the reports live only on your machine. Nobody else can view them.

This chapter is about taking the next step — moving your work from your computer to the cloud, so your reports can be accessed by your colleagues, managers, and clients from any device, at any time, from anywhere in the world.

Power BI Service (also called Power BI Online) is Microsoft's cloud-based platform for publishing, sharing, securing, and managing Power BI reports. Think of it as the online home for all your Power BI work.



Power BI Desktop vs. Power BI Service — The Key Difference

Power BI Desktop is a Windows application you install on your computer. It is used for building and designing reports.

Power BI Service is a website (app.powerbi.com) that runs in your browser. It is used for publishing, sharing, and managing reports.

You always build in Desktop first, then publish to the Service — the two tools are meant to be used together.

What You Will Learn in This Chapter

Section	Topic	What You Will Be Able to Do
1	Publishing Reports	Send your completed report from Desktop to the cloud
2	Workspaces & Sharing	Organise reports and share them with your team
3	Row-Level Security (RLS)	Control which users see which rows of data
4	Gateways	Connect your live on-premises data sources to Power BI Service
5	Scheduled Refresh	Set your data to update automatically at a set time
6	Bookmarks & Buttons	Create interactive navigation and storytelling in reports
7	App Creation	Package your reports into a professional app for end users
8	Practice Exercise	Apply all concepts in an end-to-end deployment scenario

Section 1: Publishing Reports to Power BI Service

Publishing is the process of uploading your Power BI Desktop report (.pbix file) to the cloud. Once published, your report becomes accessible through the Power BI Service website, where it can be viewed and interacted with by anyone you grant access to.

1.1 What Happens When You Publish?

When you publish a report, two things are created in the Power BI Service:

What is Created	What It Contains	How It Is Used
Report	All your pages, visuals, charts, filters, and formatting — everything the user sees	Viewed and interacted with by users
Dataset (Semantic Model)	The data, relationships, and DAX measures behind the report — the engine that powers it	Used to refresh data and build new reports

Figure 1.1 — What publishing creates in Power BI Service



An Analogy — Report vs. Dataset

Think of a Dataset as a database engine — it holds and processes the data.

Think of a Report as a window into that database — it displays the data visually.

You can build multiple reports from one dataset, just as you can look through multiple windows into the same room.

1.2 Prerequisites Before Publishing

Before you can publish, make sure the following are in place:

- You have a Power BI account — either a free account or a Pro/Premium licence.
- You are signed into Power BI Desktop with your Microsoft work or school account.
- Your report (.pbix file) is saved and has no critical errors.
- You have decided which Workspace to publish to (covered in Section 2).



Important

A free Power BI licence allows you to publish, but you cannot share reports with others unless they also have a Pro licence or the workspace is in a Premium capacity.

For classroom and learning purposes, your instructor will have set up the appropriate licences for your account.

1.3 Step-by-Step: How to Publish a Report

Follow these steps to publish your first report from Power BI Desktop to Power BI Service:

1

Sign In to Power BI Desktop

Open Power BI Desktop on your computer.

In the top-right corner, click 'Sign in' and enter your Microsoft work or school email address.

Enter your password and complete any multi-factor authentication if prompted.

Once signed in, your name or initials will appear in the top-right corner.

2

Open Your Report File

Go to File → Open Report and browse to your .pbix file, OR

Simply double-click the .pbix file from Windows Explorer to open it in Power BI Desktop.

Confirm that all your visuals load correctly and there are no error messages on any page.

3

Click Publish

In the Home ribbon at the top, locate the 'Publish' button on the right side of the ribbon.

Click the Publish button.

A dialogue box will appear asking you to choose a destination workspace.

4

Choose a Destination Workspace

A window titled 'Publish to Power BI' will appear, showing a list of available workspaces.

Select the workspace where you want your report to be published.

For now, you can use 'My Workspace' (your personal space) for practice.

Click 'Select' to confirm.

5

Wait for the Upload to Complete

A progress indicator will appear. The time taken depends on your report size and internet speed.

Once complete, a success message will appear: 'Publishing to Power BI — Success!'

A clickable link — 'Open [Report Name] in Power BI' — will appear in the dialogue.

Click the link to open your report directly in the browser.

1.4 Viewing Your Published Report in Power BI Service

After clicking the link, your browser will open Power BI Service (app.powerbi.com) and display your report. This is the same report you built in Desktop, now running in the cloud.



What Has Changed in the Service?

Your report looks and behaves the same as it did in Desktop — same pages, same visuals, same slicers.

You can interact with the report (click visuals, use slicers) but you cannot edit the layout from the Service unless you use the browser-based editing mode.

The data is a static snapshot of what was in your .pbix file at the time of publishing. To get fresh data, you will need a data refresh (covered in Section 5).

1.5 Re-Publishing an Updated Report

If you make changes to your report in Desktop and want to update the version in the Service, simply click Publish again. You will be asked if you want to replace the existing dataset and report.



Important

When you re-publish, the existing Report in the Service is replaced. Any edits made directly in the Service (such as layout changes) will be overwritten.

The Dataset is also replaced — this means any scheduled refreshes or RLS settings you configured in the Service will remain, but the data model structure will be updated.

Always keep your .pbix file as the master copy. Make all structural changes in Desktop.

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Section 2: Workspaces and Sharing

Now that your report is published to the cloud, the next step is understanding how to organise it properly and share it with the right people. This is where Workspaces come in.

2.1 What Is a Workspace?

Workspace	A shared container in Power BI Service where reports, datasets, dashboards, and dataflows are stored. Think of it as a shared project folder in the cloud that a team can access together.
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Power BI Service has two types of workspaces:

Type	Description	Best Used For
My Workspace	A personal, private workspace that only you can access. It is created automatically for every user.	Personal drafts, testing, and learning — not for team collaboration
Shared Workspace (App Workspace)	A collaborative workspace created and managed by an admin. Multiple users can be added with different roles.	Team projects, departmental reports, and production-grade content

Figure 2.1 — My Workspace vs. Shared Workspace

2.2 Creating a New Workspace

As a report author, you can create a new workspace for your team. Here is how:

1	Open Workspace Panel In Power BI Service, click 'Workspaces' in the left navigation panel. A panel will slide out showing all workspaces you are a member of.
2	Create Workspace At the bottom of the workspace panel, click '+ New workspace'. A configuration panel will open on the right side of the screen.
3	Configure the Workspace Enter a Name — e.g., 'Sales Analytics Team' or 'Finance Reports Q4'. Optionally add a Description to help team members understand what this workspace contains. Expand 'Advanced' settings if you need to configure a Premium licence or a OneDrive folder for the workspace.

Click 'Save' to create the workspace.

2.3 Workspace Roles — Who Can Do What?

Each person you add to a workspace is assigned a Role. The role determines what they can see and do in that workspace. There are four roles:

Role	What They Can Do	Typical User
Admin	Full control — add/remove members, delete workspace, publish, edit everything	Workspace owner, IT administrator
Member	Publish and edit content, manage data sources, but cannot add/remove members	Senior analyst, report author
Contributor	Publish and edit reports and datasets, but cannot change workspace settings or manage access	Data analyst, report developer
Viewer	View and interact with reports and dashboards only — cannot edit or publish	Business stakeholder, manager, end user

Figure 2.4 — Workspace Roles and Permissions

2.4 Adding Members to a Workspace

1

Open Workspace Settings

Navigate to your workspace in Power BI Service.
Click the three-dot menu (...) next to the workspace name, then select 'Workspace settings'.
OR click 'Access' in the workspace header bar.

2

Add a Person or Group

In the Access panel, type the person's email address or a security group name in the search box.
Select their name from the search results.
Choose their role from the dropdown (Admin, Member, Contributor, or Viewer).
Click 'Add' to confirm.

2.5 Sharing a Report Directly

In addition to adding users to a workspace, you can also share an individual report directly with specific people. This is useful when you want someone to view one specific report without giving them access to the entire workspace.

1

Open the Report in Power BI Service

Navigate to the report you want to share.
Click the 'Share' button in the top toolbar.

2**Configure Share Settings**

Enter the email addresses of the people you want to share with.

Choose whether recipients can re-share the report with others.

Choose whether to include a link to the underlying dataset (allows building new reports).

Click 'Send' — recipients will receive an email with a direct link to the report.

**Workspace vs. Direct Share — When to Use Which?**

Use Workspace roles when you want a team to have ongoing access to all content in a workspace and collaborate on building reports.

Use Direct Share when you want a specific person to view one report without giving them access to the whole workspace.

For large-scale distribution to many end users, use Apps (covered in Section 7) — apps are the recommended approach for production report delivery.

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Section 3: Row-Level Security (RLS)

Imagine you have a single sales report that covers all regions — North, South, East, and West. You want to share this report with all four regional managers. However, each manager should only see their own region's data — not the data of other regions.

You could create four separate reports, but that is inefficient and hard to maintain. The correct solution is Row-Level Security.

3.1 What Is Row-Level Security?

Row-Level Security (RLS)

A feature in Power BI that restricts which rows of data each user can see. All users view the same report, but DAX filters are applied dynamically based on the user's identity, so each person only sees the data they are authorised to see.



A Real-World Example

A national retail company has 50 store managers. Each manager should see only their own store's sales when they open the dashboard.

With RLS, one report is built and shared with all 50 managers. Power BI automatically filters the data to show each manager only their own store's records.

The IT team maintains one report instead of 50 — and adding a new manager is just adding one row to a mapping table.

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3.2 How RLS Works — The Concept

RLS works in two parts:

1. Define Roles — In Power BI Desktop, you create named roles and write DAX filter expressions that determine which rows each role can see.
2. Assign Users to Roles — In Power BI Service, you assign users (by email) to those roles. When that user logs in and opens the report, their role's filter is applied automatically.

Where You Do It	What You Do
Power BI Desktop (before publishing)	Create roles and write DAX filter expressions
Power BI Service (after publishing)	Assign user email addresses to roles

Figure 3.1 — RLS is set up in Desktop, then managed in Service

3.3 Setting Up RLS in Power BI Desktop

Step 1 — Open the Manage Roles Interface

In Power BI Desktop, go to the Modeling ribbon. Click 'Manage Roles'.

Step 2 — Create a New Role

In the Manage Roles window, click 'Create'. Type a name for the role. Use a descriptive name — e.g., 'North Region', 'South Region', or 'Store Manager'.

Step 3 — Write the DAX Filter Expression

For each role, you write a DAX expression that evaluates to TRUE or FALSE for every row in a table. Only rows where the expression returns TRUE will be visible to that role.

Example — Creating a role that shows only North Region data:

DAX Filter

-- Filter on the Region column of the Sales table:

[Region] = "North"

-- For a dynamic filter using the logged-in user's email:

[Manager Email] = USERPRINCIPALNAME()

Figure 3.4 — DAX filter expressions for RLS roles



USERPRINCIPALNAME() — The Dynamic RLS Function

USERPRINCIPALNAME() is a DAX function that returns the email address of the currently logged-in user.

When you use this in an RLS filter, Power BI compares the logged-in user's email to a column in your data (e.g., a Manager Email column).

This means you do not need to create a separate role for each individual user — one role handles all of them dynamically.

Example: [Manager Email] = USERPRINCIPALNAME() — this shows each manager only the rows where their own email appears.

Step 4 — Test Your RLS in Desktop

Before publishing, test the roles to make sure the filters are working correctly.

In the Modeling ribbon, click 'View as'. Select the role you want to test. Your report will now show you what a member of that role would see.

3.4 Assigning Users to Roles in Power BI Service

After publishing your report, go to Power BI Service and assign real users to the roles you created.

1

Navigate to the Dataset

In your workspace, find the Dataset (Semantic Model) associated with your report.

Click the three-dot menu (...) next to the dataset name.
Select 'Security'.

2

Assign Users to Roles

The Row-Level Security page will list all the roles you created in Desktop.
Click on a role (e.g., 'North Region').
In the text box, type the email address of the person to assign to this role.
Click 'Add' and then 'Save'.

✓ RLS Setup Complete

Once users are assigned, they will only see data filtered by their role — even though they are viewing the same report as everyone else.
Changes to role membership take effect immediately — no need to re-publish.
You can assign multiple users to the same role, and a user can be assigned to multiple roles.

3.5 Static vs. Dynamic RLS — When to Use Which

Static RLS	Dynamic RLS
One role per group/region/category	One role for all users
Filter value is hardcoded: [Region] = "North"	Filter uses the login email: [Email] = USERPRINCIPALNAME()
Must create and assign a new role each time a new region/group is added	Adding a new user just means adding a row to the mapping table in data
Best for a small, fixed number of categories (e.g., 4 regions)	Best for large user bases where memberships change frequently (e.g., 200 salespeople)

Figure 3.8 — Static vs. Dynamic RLS Comparison

Section 4: Gateways — Connecting On-Premises Data to the Cloud

Many organisations store their data in systems that are inside their own office network — not on the internet. This includes databases on office servers, Excel files on shared drives, or enterprise systems like SAP. This type of data is called on-premises data.

Power BI Service lives in the cloud (Microsoft's data centres). By default, it cannot access your office's internal systems directly. A Gateway solves this problem.

4.1 What Is a Gateway?

On-Premises Data Gateway	A software bridge installed on a computer inside your office network. It creates a secure, encrypted tunnel between your internal data sources and Power BI Service, allowing cloud-based Power BI to read and refresh your on-premises data.
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An Analogy — The Gateway as a Secure Courier

Imagine Power BI Service is your office in Mumbai and your database is in a secure facility in Bengaluru. You cannot directly access the Bengaluru facility from Mumbai — it is behind locked gates. The Gateway is like a trusted courier who has a key to both locations. Power BI sends a request to the Gateway, the Gateway goes inside the facility, retrieves the data, and securely delivers it back to Power BI. Your internal data never becomes publicly accessible — it always travels through the secure, encrypted gateway channel.

4.2 Types of Gateways

Gateway Type	Description	Best Used For
Standard (Enterprise) Mode	Installed on a dedicated computer that is always on. Supports multiple users and multiple data sources. Can be managed centrally.	Production environments, team-wide data refreshes, multiple reports accessing the same source
Personal Mode	Installed on your own personal computer. Only you can use it. The computer must be turned on for refreshes to work.	Individual use, development and testing, small-scale scenarios

Figure 4.1 — Standard Gateway vs. Personal Mode Gateway

4.3 When Do You Need a Gateway?

You need a gateway only when your data source is on-premises. If your data is already in the cloud (e.g., Azure SQL Database, SharePoint Online, Salesforce), no gateway is required.

Data Source	Gateway Required?	Reason
SQL Server on office server	Yes	Lives inside your internal network
Excel file on a local computer	Yes	File is not accessible from the internet
Excel file on SharePoint Online	No	Already in the Microsoft cloud
Azure SQL Database	No	Already in the Microsoft cloud
Oracle / SAP on-premises	Yes	Lives inside your internal network
Google BigQuery	No	Cloud-based service

Figure 4.2 — When a gateway is and is not required

4.4 Installing the Gateway (Overview)

The gateway is downloaded and installed by an IT administrator on a dedicated machine inside the office network. Here is an overview of the process:

1

Download the Gateway Installer

In Power BI Service, go to Settings (gear icon, top right) → Manage gateways.
On the Gateways page, click '+ New' or download the gateway installer from the link provided.
The installer file (.exe) will download to your computer.

2

Install on the Gateway Machine

Run the downloaded installer on the machine that will act as the gateway (must be on the office network).
Sign in with your Power BI account during installation.
Choose 'On-premises data gateway (recommended)' for the Standard mode.
Give the gateway a descriptive name (e.g., 'Finance-Server-Gateway').
The gateway will register itself with Power BI Service and appear in your Manage Gateways page.

3

Add a Data Source to the Gateway

In Power BI Service, go to Settings → Manage gateways.
Select your gateway, then click '+ Add data source'.
Choose the data source type (e.g., SQL Server, Oracle, File).
Enter the server name, database name, and credentials.
Click 'Add' — the data source is now accessible through the gateway.



Note

The gateway machine must be turned on and connected to the internet for scheduled refreshes to work.
For enterprise deployments, install the gateway on a dedicated server, not on a personal laptop.
Gateway installation and management is typically performed by an IT administrator, not an end user.

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Section 5: Scheduled Refresh

When you publish a report, the data it contains is a snapshot — it reflects whatever was in your data source at the time you last refreshed and published. Without a refresh schedule, your report will show the same data forever.

Scheduled Refresh solves this by automatically fetching fresh data from your data source at times you specify — so your stakeholders always see up-to-date information when they open the report.

5.1 What Happens During a Refresh?

During a scheduled refresh, Power BI Service:

3. Connects to your data source using the gateway (if on-premises) or a direct cloud connection.
4. Runs the same queries that Power BI Desktop ran when you first connected to the data.
5. Loads the new data into the published dataset (Semantic Model).
6. All reports and dashboards built from that dataset now display the refreshed data — automatically.



Key Point — Refresh Updates the Dataset, Not the Report File

The .pbix file on your computer is NOT updated by a scheduled refresh.

Only the Dataset in Power BI Service is refreshed.

Your report pages and visuals are not changed — they simply draw from the freshly updated dataset.

If you need to change the report layout or structure, you must do that in Desktop and re-publish.

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5.2 Prerequisites for Scheduled Refresh

Requirement	Detail
Dataset must be in Power BI Service	You must have published your report before setting up a refresh schedule
Data source credentials configured	Power BI needs stored credentials to connect to your data source
Gateway configured (for on-premises data)	If your data is on-premises, the gateway must be installed and the data source added
Pro or Premium licence	Scheduled refresh is not available on free accounts for shared workspaces

Figure 5.1 — Prerequisites for scheduled refresh

5.3 Configuring a Scheduled Refresh — Step by Step

1

Open Dataset Settings

In your workspace in Power BI Service, find the Dataset associated with your report.
Click the three-dot menu (...) next to the dataset name.
Select 'Settings'.

2

Configure Data Source Credentials

In Dataset Settings, find the 'Data source credentials' section.
Click 'Edit credentials' next to each data source listed.
Enter the username and password (or other authentication method) for that source.
Click 'Sign in' or 'Save'.
This tells Power BI how to log into your database or file source during the refresh.

3

Turn On Scheduled Refresh

Scroll down to the 'Scheduled refresh' section in Dataset Settings.
Toggle the switch to 'On'.
Select the Refresh frequency: Daily or Weekly.
Set the Time Zone to match your local time zone.
Click '+ Add another time' to add specific times for the refresh to run.
You can add up to 8 refresh times per day (48 with Premium capacity).

4

Save the Configuration

Click 'Apply' to save your refresh schedule.
Power BI will now automatically refresh your dataset at the times you specified.
You will receive an email notification if a refresh fails.



Pro Tip

Schedule refreshes during off-peak hours (e.g., early morning) to avoid impacting system performance during business hours.

Always set up failure notifications — this alerts you if a refresh fails so you can investigate before users notice stale data.

You can also trigger a manual refresh at any time by clicking the refresh icon next to the dataset in the workspace.

5.4 Monitoring Refresh History

Power BI keeps a log of all refresh attempts so you can check whether refreshes are succeeding or failing.

In the Dataset's three-dot menu, select 'Refresh history'. This shows a table of all past refresh attempts with their status (Completed or Failed), start time, and duration.

Status	What It Means	What to Do
Completed	Refresh ran successfully and data was updated	No action required
Failed	Refresh did not complete — data may be stale	Click the error icon for details, check credentials and gateway status
In Progress	Refresh is currently running	Wait for it to complete before checking reports
Cancelled	Refresh was manually cancelled or timed out	Trigger a manual refresh after investigating the cause

Figure 5.6 — Understanding refresh status indicators

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Section 6: Bookmarks and Buttons

One of the most powerful ways to make your Power BI reports more professional and user-friendly is to add interactive navigation. Instead of presenting raw charts and tables, you can guide users through your report like a presentation — using Bookmarks and Buttons.

6.1 What Is a Bookmark?

Bookmark

A saved snapshot of the current state of a report page — including which visuals are visible, what filters are applied, and what the current slicer selections are. You can use bookmarks to create navigation, storytelling, and toggle effects.

What Does a Bookmark Capture?

- Which visuals are visible (shown or hidden on the page)
- Current filter context (what filters are active in the Filters pane)
- Current slicer selections (what values are selected in each slicer)
- Visual-level drill state (what level of a hierarchy is being shown)
- Page navigation (which page the report is on)

6.2 Creating a Bookmark

1

Set Up Your Report State

Make sure your report page looks exactly how you want the bookmark to capture it. Apply any filters, set slicer values, hide or show visuals as needed.

2

Open the Bookmarks Pane

In Power BI Desktop, go to View ribbon → check 'Bookmarks Pane'. The Bookmarks pane will appear on the right side of the screen.

3

Add the Bookmark

In the Bookmarks pane, click '+ Add'.
A new bookmark appears, named 'Bookmark 1' by default.
Double-click the name to rename it meaningfully — e.g., 'Revenue View', 'Filter: North Region', or 'Chart Hidden'.

6.3 What Are Buttons?

Button

A clickable element you can add to a Power BI report page. When a user clicks a button, it can trigger a bookmark, navigate to another page, open a URL, or perform other actions.

Types of Buttons in Power BI

Button Type	What It Does	Common Use Case
Blank	A plain button with custom text and formatting	Custom navigation or toggle actions
Back	Navigates to the previous page viewed	Back navigation in a drillthrough scenario
Bookmark	Triggers a specific saved bookmark	Switching between views or showing/hiding visuals
Page Navigation	Jumps to a specific report page	Creating a custom navigation menu
Drillthrough	Opens the drillthrough page for the selected context	Detailed view for a selected item
Q&A	Opens the natural language Q&A interface	Allowing users to ask questions in plain language
URL	Opens a web link in a browser	Linking to external documentation or websites

Figure 6.3 — Types of buttons available in Power BI Desktop

6.4 Adding and Configuring a Button

1

Insert a Button

Go to Insert ribbon in Power BI Desktop.

Click 'Buttons' and choose the button type (e.g., Blank or Bookmark).

The button will appear on your report canvas. Drag it to the desired position.

2

Customise the Button Appearance

With the button selected, use the Format pane (paint roller icon) on the right.

Under 'Button', set the Style (fill colour, border, text).

Under 'Button text', type the label you want (e.g., 'Show Chart', 'North Region', or 'Back').

Set different styles for Default state (normal) and On Hover state (when mouse is over it).

3

Assign an Action to the Button

In the Format pane, scroll to the 'Action' section.

Toggle 'Action' to On.

Set Type to 'Bookmark' (or Page navigation, URL, etc.).

If Bookmark is selected, choose the bookmark from the dropdown list.

**Pro Tip**

To test your buttons in Desktop, hold Ctrl and click the button — this simulates the button click as it would work in the Service.

Use buttons and bookmarks together to create 'show/hide' toggle panels — for example, a button that shows a help overlay or a filter panel.

Consistent button styling across a report gives it a professional, app-like feel. Create a colour and font standard and stick to it.

6.5 Practical Example — Creating a View Toggle

A common and highly useful pattern is creating a toggle between two views of the same data — for example, a chart view and a table view.

1

Prepare Two Visual States

Place a bar chart and a table visual on the same report page, showing the same data.

Position them on top of each other (they will overlap — only one will be visible at a time).

2

Create Two Bookmarks

Show only the bar chart (hide the table using the Selection pane). Add a bookmark named 'Chart View'.

Show only the table (hide the bar chart). Add a bookmark named 'Table View'.

3

Add Two Toggle Buttons

Add a button labelled 'Show Chart' — assign action: bookmark 'Chart View'.

Add a button labelled 'Show Table' — assign action: bookmark 'Table View'.

Style the active button differently to show which view is currently selected.

Section 7: App Creation

So far, you have been working with reports and workspaces — tools that are primarily used by report authors and analysts. But what about the end users — the managers, executives, and business teams who simply want to view and interact with the data?

Power BI Apps are designed for exactly this purpose. An App is a polished, packaged collection of reports and dashboards that is published to end users in a structured, easy-to-navigate format.

7.1 What Is a Power BI App?

Power BI App

A published collection of related reports and dashboards, organised into a navigation structure and distributed to end users as a standalone product. Apps separate the authoring experience (workspace) from the consuming experience (app).

 Why Use Apps Instead of Sharing Workspaces?

When you give someone access to a workspace, they can see all reports, datasets, and files — including works-in-progress and draft versions.

An App shows users only the finalised, approved content — in a clean, structured interface that you control.

Apps can be updated (by publishing a new version) without the end user noticing any disruption.

Apps can be distributed to a large number of users at once, without sharing each report individually.

7.2 App vs. Workspace — The Key Distinction

Workspace (Behind the Scenes)	App (What End Users See)
For report authors and collaborators	For end users and report consumers
Contains all content including drafts and works-in-progress	Contains only the published, approved, final content
Shows raw datasets, reports, dataflows, and dashboards	Shows only the reports and dashboards you choose to include
Accessed via Power BI Service workspace navigation	Accessed via the Apps section in Power BI Service or a direct link
Changes are immediately visible to all workspace members	Changes only appear when the author publishes a new version of the App

Figure 7.1 — Workspace vs. App: Author experience vs. Consumer experience

7.3 Creating a Power BI App — Step by Step

1

Navigate to Your Workspace

In Power BI Service, open the workspace that contains the reports you want to include in the app.

Make sure all the reports you want to publish are finalised and error-free.

2

Click 'Create App'

In the workspace, click the 'Create app' button in the top toolbar.
The App editor will open — a dedicated interface for designing your app.

3

Configure the Setup Tab

In the Setup tab, enter the App name (this is what users will see).
Write a Description to help users understand what the app is about.
Upload an App logo (optional but recommended for a professional look).
Choose a Theme colour for the app navigation bar.

4

Build the Navigation in the Content Tab

Click the 'Content' tab in the App editor.
Click '+ Add content' to add reports and dashboards from your workspace.
Drag the items to reorder them — this determines the order in the app's navigation.
Add sections and separators to organise related reports into groups.
Each item in the navigation can be renamed for clarity (e.g., rename a report called 'PBI_Sales_Q4_FINAL_v3' to 'Q4 Sales Report').

5

Set Audience and Permissions in the Audience Tab

Click the 'Audience' tab.
By default, there is one audience group called 'All users in the organisation' — this shares the app with everyone.
Click '+ New audience group' to create different versions for different groups.
For each audience group, add specific users, security groups, or distribution lists.
You can also customise which reports each audience group can see.

6

Publish the App

Once all tabs are configured, click 'Publish app' in the top right corner.
A confirmation dialogue will appear — click 'Publish'.
The app is now live. Users who have been given access can find it in the 'Apps' section of Power BI Service.
Share the direct link with users or instruct them to install the app from the Power BI Apps marketplace.

7.4 Updating an Existing App

When you update reports in your workspace and want to push those updates to app users, you need to re-publish the app.

- 7. Make your changes to the reports in the workspace (or re-publish from Desktop).
- 8. Go to the workspace and click 'Update app' (the same button that previously said 'Create app').
- 9. Review the configuration and click 'Update app' to confirm.



Note

App users do not need to do anything — the updated app is available immediately after you publish.
The app URL stays the same — users do not need a new link.
Changes are instantly reflected when users next open the app or refresh their browser.

7.5 The End-User App Experience

When an end user opens the app, they see a clean, structured navigation experience — not the cluttered workspace view that authors use.

What the Author Sees (Workspace)	What the End User Sees (App)
All reports, datasets, dataflows, dashboards — including drafts	Only the final, published reports you chose to include
Technical file names (e.g., Sales_FINAL_v2_Oct)	Friendly display names you assigned (e.g., 'October Sales Report')
Edit buttons, publish options, dataset settings	Clean reading and interaction interface — no editing controls
All workspace members listed in the side panel	Navigation sections and reports only

Figure 7.8 — Author experience vs. End-user experience comparison

Section 8: Practice Exercise — End-to-End Deployment

This exercise brings together all the concepts from this chapter. You will take a completed Power BI Desktop report and deploy it fully using Power BI Service — from publishing to securing to distributing it as an App.

8.1 Business Scenario

You are a data analyst at Meridian Electronics, a mid-sized electronics retailer with four regional offices: North, South, East, and West. The management team has approved your Sales Performance Dashboard for company-wide distribution.

You have been asked to:

10. Publish the report to the company's Power BI workspace.
11. Set up Row-Level Security so each regional manager sees only their region.
12. Configure a daily scheduled refresh so data is updated every morning.
13. Add bookmark-based navigation to allow users to switch between summary and detail views.
14. Package everything into a Power BI App for distribution to the sales team.

8.2 Exercise Step-by-Step Guide

Part A — Publish the Report

1

Open the report in Power BI Desktop

Make sure the file is saved and all visuals display correctly.

2

Sign in and publish

Sign in with your Power BI account.

Click Home → Publish.

Choose the 'Meridian Sales Team' workspace as the destination.

Wait for the success message and click the link to confirm the report is visible in the Service.

Part B — Configure RLS

3

Create roles in Desktop

Go to Modeling → Manage Roles.

Create four roles: North Manager, South Manager, East Manager, West Manager.

For each role, apply the filter: Sales[Region] = "<Region Name>" (replace with the actual region).

Test each role using 'View as Roles' to confirm the data is filtered correctly.

Save and re-publish the report to the Service.

4

Assign users to roles in Service

In Power BI Service, go to the Dataset → Security.

Assign each regional manager's email to their corresponding role.

Verify by asking one manager to open the report and confirm they see only their region.

Part C — Set Up Scheduled Refresh

5

Configure credentials and refresh

In the workspace, open the dataset Settings.

Under Data source credentials, enter the database credentials.

Under Scheduled refresh, turn on the schedule.

Set to Daily, at 6:00 AM IST.

Click Apply and confirm the schedule is saved.

Part D — Add Bookmarks and Buttons

6

Create a view toggle

In Power BI Desktop, create two bookmarks: 'Summary View' (shows KPI cards and chart) and 'Detail View' (shows the data table).

Add two buttons labelled 'Summary' and 'Details'.

Assign the corresponding bookmarks as actions.

Re-publish to the Service.

Part E — Create the App

7

Build and publish the app

In the workspace, click 'Create app'.

Name: 'Meridian Sales Performance App'.

Add the Sales Dashboard and any related reports to the Content tab.

Rename navigation items to be user-friendly.

In the Audience tab, add the sales team distribution list.

Click Publish app and share the URL with the team.

8.3 Expected Outcomes Checklist

Task	Expected Outcome	Status
Report Published	Report visible in 'Meridian Sales Team' workspace in Power BI Service	[]
RLS Configured	Each regional manager sees only their own region's data	[]
Credentials Set	No credential errors in Dataset Settings	[]
Refresh Scheduled	Refresh history shows 'Completed' for the 6 AM slot	[]
Bookmarks Created	Two bookmarks saved — Summary View and Detail View	[]
Buttons Functional	Ctrl + Click on buttons in Desktop triggers correct bookmarks	[]
App Published	App accessible via the URL — navigation and content correct	[]
User Access Verified	A team member confirms they can open the app and see their data	[]

Figure 8.1 — Exercise completion checklist

8.4 Common Mistakes to Avoid

Mistake	Why It Causes a Problem	How to Avoid It
Publishing to My Workspace instead of the team workspace	Others cannot access content in My Workspace	Always check the destination workspace before clicking Select
Forgetting to re-publish after changing RLS in Desktop	Old roles remain in the Service — new roles are not applied	After any Desktop change, always re-publish to keep Service in sync
Not setting data source credentials in the Service	Scheduled refresh fails with a credentials error	Always configure credentials in Dataset Settings immediately after publishing
Using the same bookmark name twice	Power BI may apply the wrong bookmark to a button	Use unique, descriptive names for all bookmarks
Adding unfinished reports to the app	End users see draft or error-containing reports	Only add reports to the app after they have been fully tested and approved

Figure 8.2 — Common deployment mistakes and how to avoid them

Chapter Summary

In this chapter, you have moved beyond report building and learned how to deploy, secure, and distribute Power BI content professionally. Here is a recap of all the key topics covered:

Section	Topic	Key Concept
1	Publishing Reports	Use Home → Publish in Desktop to upload reports and datasets to Power BI Service
2	Workspaces & Sharing	Organise content in team workspaces; assign roles (Admin, Member, Contributor, Viewer)
3	Row-Level Security	Define roles in Desktop with DAX filters; assign users to roles in Service
4	Gateways	Install a gateway on an office machine to connect on-premises data to the cloud
5	Scheduled Refresh	Configure credentials and a time schedule so datasets update automatically
6	Bookmarks & Buttons	Save visual states as bookmarks; attach them to buttons for interactive navigation
7	App Creation	Package reports into a clean, structured app and distribute to end users

Figure Summary — Advanced Features & Power BI Service — Chapter at a Glance



What Comes Next?

In the next chapter, you will explore Power BI Premium features — including paginated reports, AI-powered visuals (Key Influencers, Decomposition Tree), and integration with Microsoft Teams.

You will also learn about Power BI Embedded — how to integrate Power BI reports directly into custom web applications.

These topics build directly on the deployment and sharing knowledge you have built in this chapter.